

Anas MADANI

Full Stack Software Engineer

anasmadani49@gmail.com | +212 6 66 32 24 50 | Casablanca, Morocco
linkedin.com/in/anas-madani | github.com/HexNebula



PROFILE

Software engineer graduated from ENSIAS (Software Engineering), full stack from backend to frontend, with a DevOps practice of deploying on Linux: continuous integration and delivery, containerization and orchestration. Experience in demanding environments (banking, datacenter), used to delivering tested, reviewed and documented code within an agile team.

EDUCATION

- **École Nationale Supérieure d'Informatique et d'Analyse des Systèmes (ENSIAS)** Rabat, Morocco
Engineering Degree – Software Engineering 2022 – 2025
- **CPGE – Settati** Settati, Morocco
Preparatory Classes for Engineering Schools, TSI 2019 – 2022

PROFESSIONAL EXPERIENCE

- **Banque Centrale Populaire (BCP)** Casablanca, Morocco
Software Engineer Intern (Graduation Project) February 2025 – July 2025
 - Independently designed and delivered a banking product recommendation platform: 8 Spring Boot microservices, a FastAPI recommendation API and a React/TypeScript frontend, deployed internally and presented to business teams.
 - Built the recommendation engine end-to-end: prepared 17,000 customer-product pairs, customer segmentation via K-Means, benchmarked 4 classification algorithms against a graph-based collaborative filtering approach (Neo4j GDS), with a Precision@K / Recall@K / NDCG@K evaluation framework.
 - Industrialized the platform: Docker containerization, event-driven communication with Kafka, Redis caching, Keycloak authentication with centralized JWT validation at the API Gateway level.
 - *Stack:* Java, Spring Boot/Cloud, React, TypeScript, Python, FastAPI, Neo4j, PostgreSQL, Redis, Kafka, Keycloak, Docker, Git
- **N+ONE DATACENTERS** Casablanca, Morocco
Software Engineer Intern August 2024 – September 2024
 - Built a tender/RFP management platform (Django REST Framework / React), covering the full consultation lifecycle, candidate submissions and a public portal for external companies.
 - Implemented security mechanisms: AES-256-GCM encryption for sensitive documents, role-based access control, JWT authentication with token rotation, operation audit logging and rate limiting on public endpoints.
 - Containerized the stack with Docker Compose (Django, PostgreSQL, React, Nginx reverse proxy), set up a GitHub Actions CI pipeline (flake8, pytest against PostgreSQL) and a scheduled deadline-monitoring service.
 - *Stack:* Python, Django REST Framework, React, Vite, PostgreSQL, Docker, Nginx, GitHub Actions, JWT, AES-GCM, pytest
- **Commune Er-rich** Er-rich, Morocco
Software Engineer Intern July 2023 – August 2023
 - Independently built a Java/JavaFX desktop application for managing the municipality's workforce: profile management, absence tracking and automated payroll calculation.
 - Designed the relational database (MySQL/MariaDB): data dictionary, functional dependencies and conceptual model, based on requirements gathered from the relevant department.
 - *Stack:* Java, JavaFX, MySQL/MariaDB, phpMyAdmin, NetBeans, UML, Git

PROJECTS

- **BusConnect:** Bus reservation platform on a microservices architecture with Spring Boot 3, Spring Cloud (Eureka, API Gateway), React 18 and Kubernetes deployment. Integrates PostgreSQL, MySQL and MongoDB depending on the service. [GitHub](#)
- **MLOps Pipeline Generator:** MLOps code generator based on Model-Driven Engineering (MDE) using Eclipse Ecore and Aceleo. Automatically generates Python scripts, Dockerfiles and Kubernetes configurations from metamodels via an SWT/EMF Forms interface. [GitHub](#)
- **BacSurveillance:** Exam supervision management platform using a greedy + backtracking algorithm for automatic teacher assignment. FastAPI/PostgreSQL backend, React 19 (TypeScript) frontend, Word/Excel export and bilingual Arabic/French interface. [GitHub](#)
- **FitHub:** Full-stack gym management application on a microservices architecture with CI/CD pipelines via GitHub Actions. [GitHub](#)

SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, C, C++, SQL
Backend: Spring Boot, Spring Cloud, FastAPI, Django REST Framework
Frontend: React, TypeScript, Tailwind CSS, Vite
Databases: PostgreSQL, MySQL, MongoDB, Neo4j, Redis
DevOps & Infra: Docker, Kubernetes, Kafka, GitHub Actions, Nginx, Keycloak, Linux
Spoken Languages: Arabic (native), French (fluent), English (fluent)